

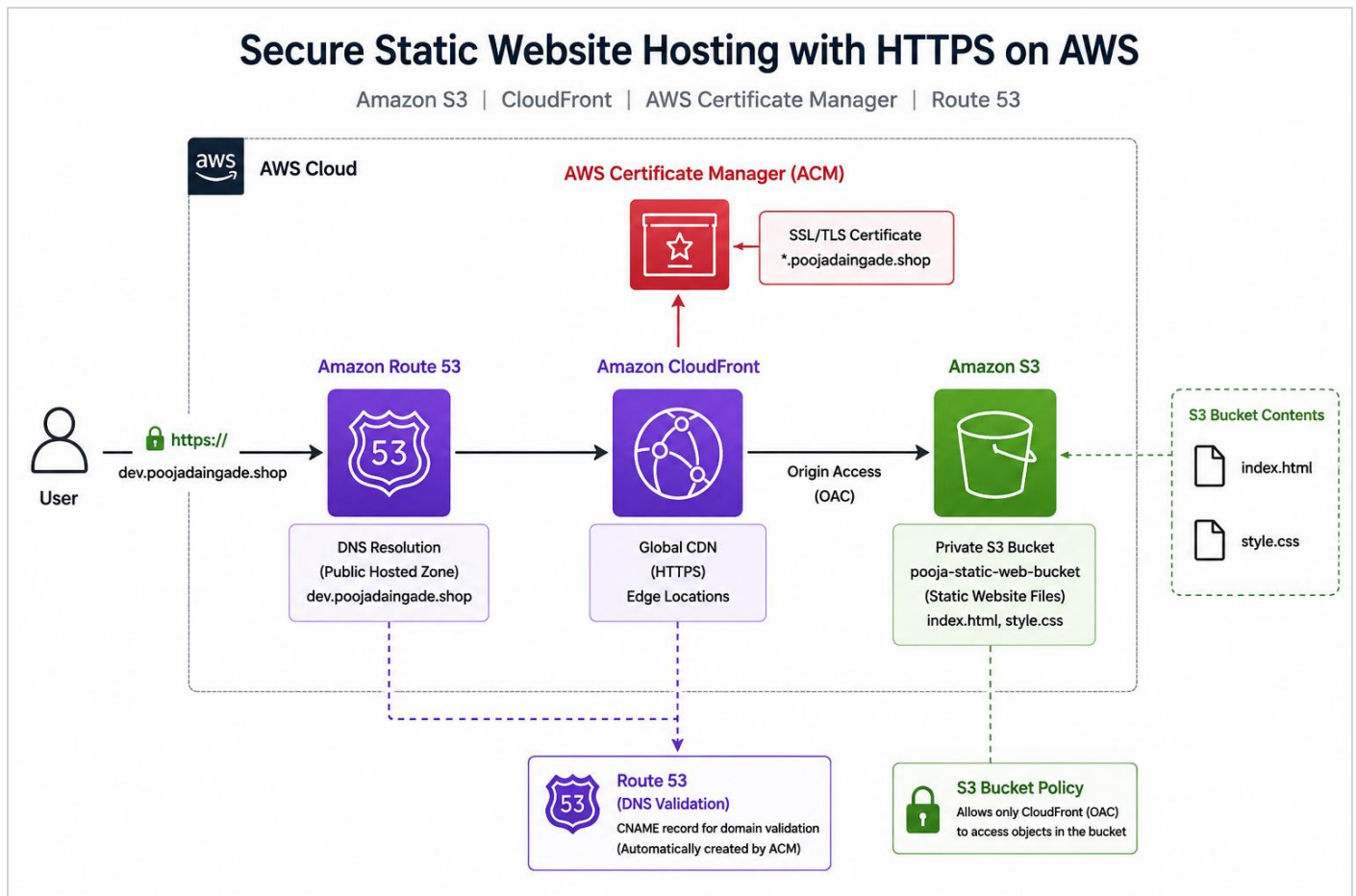
Secure Static Website Hosting with HTTPS on AWS

Prepared By: Pooja Daingade

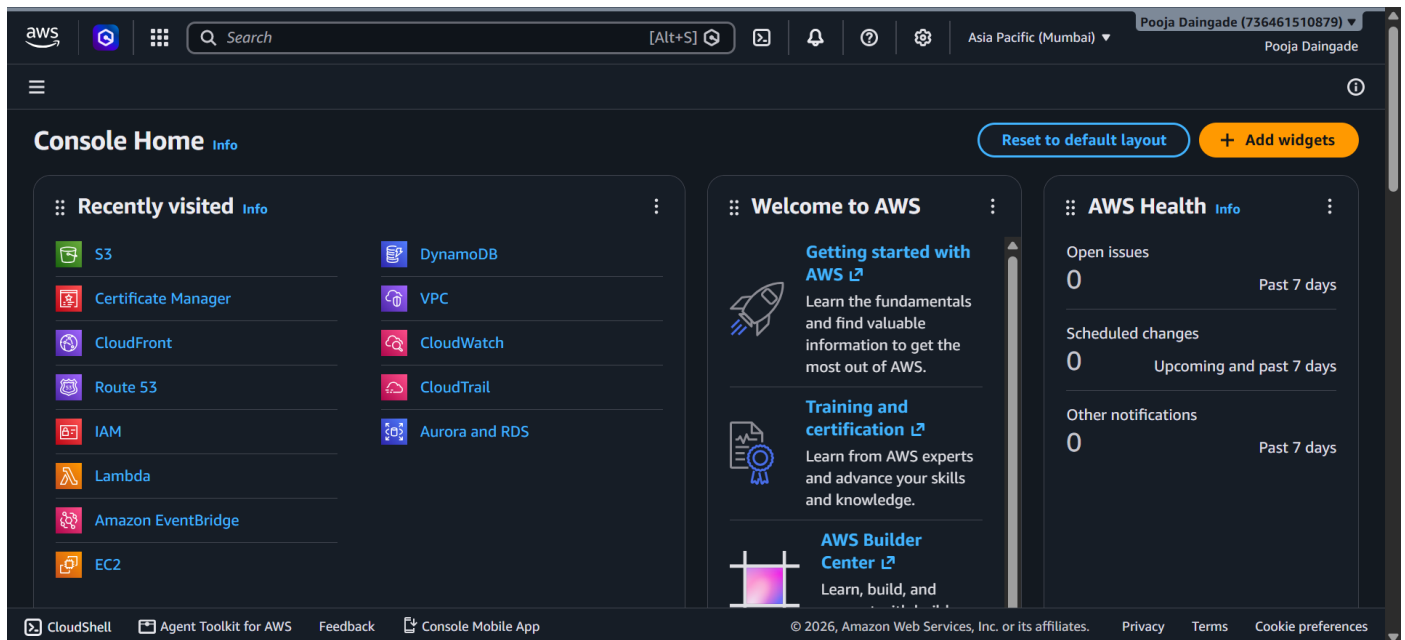
Project Overview

This project demonstrates the deployment of a secure static website on AWS using Amazon S3, CloudFront, AWS Certificate Manager (ACM), and Route 53. The website files are stored in a private S3 bucket, while CloudFront delivers the content globally over HTTPS using an SSL/TLS certificate. Route 53 is configured to route the custom domain to CloudFront, providing secure and reliable access to the website.

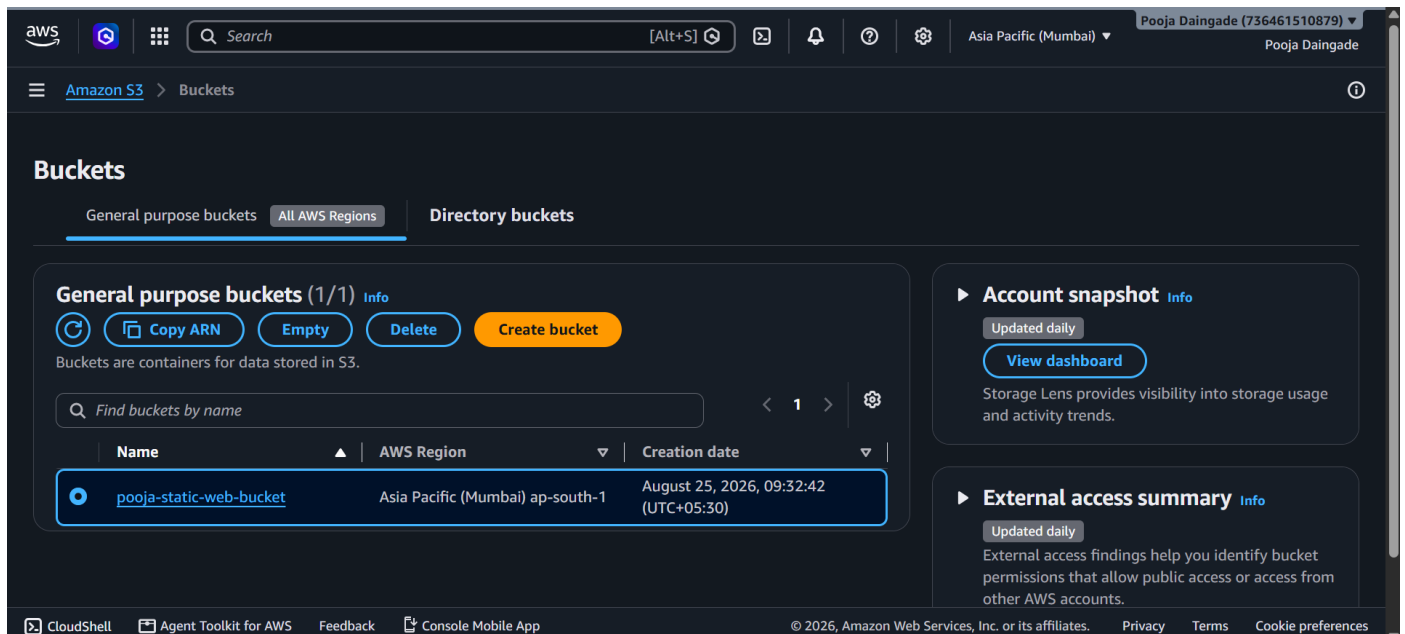
Architecture Diagram



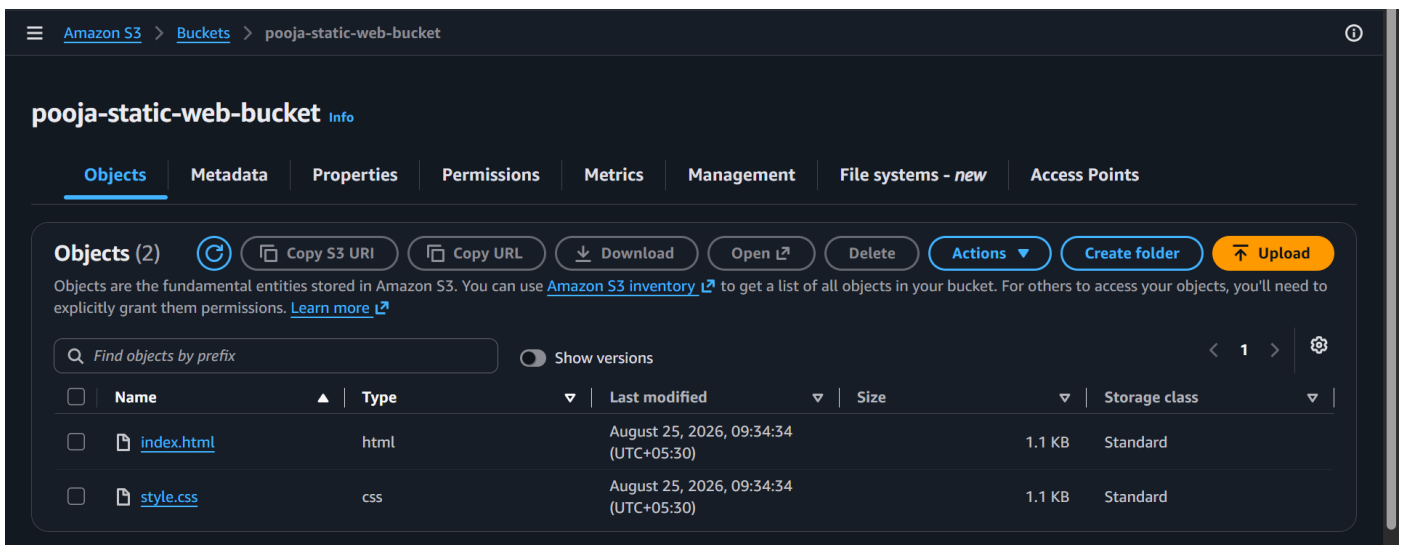
1. Logged in to the AWS Management Console and accessed the required AWS services for the project, including Amazon S3, AWS Certificate Manager (ACM), Amazon CloudFront, and Amazon Route 53.



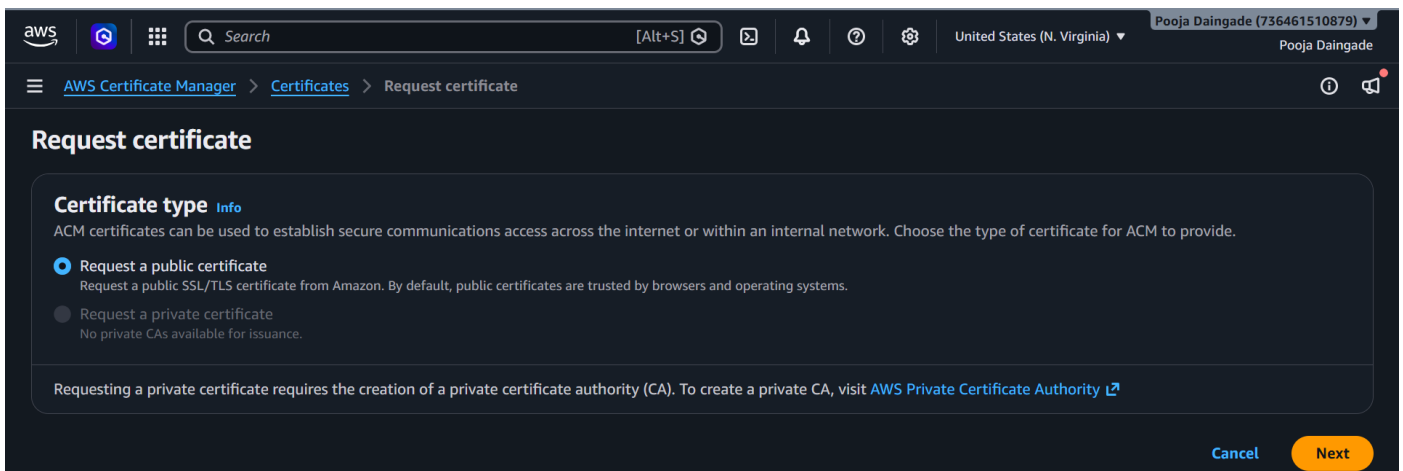
2. Created an Amazon S3 bucket named **pooja-static-web-bucket** in the ap-south-1 (Mumbai) region with **Bucket Versioning enabled** and **Block all public access enabled**. Other bucket settings were kept at their default values.



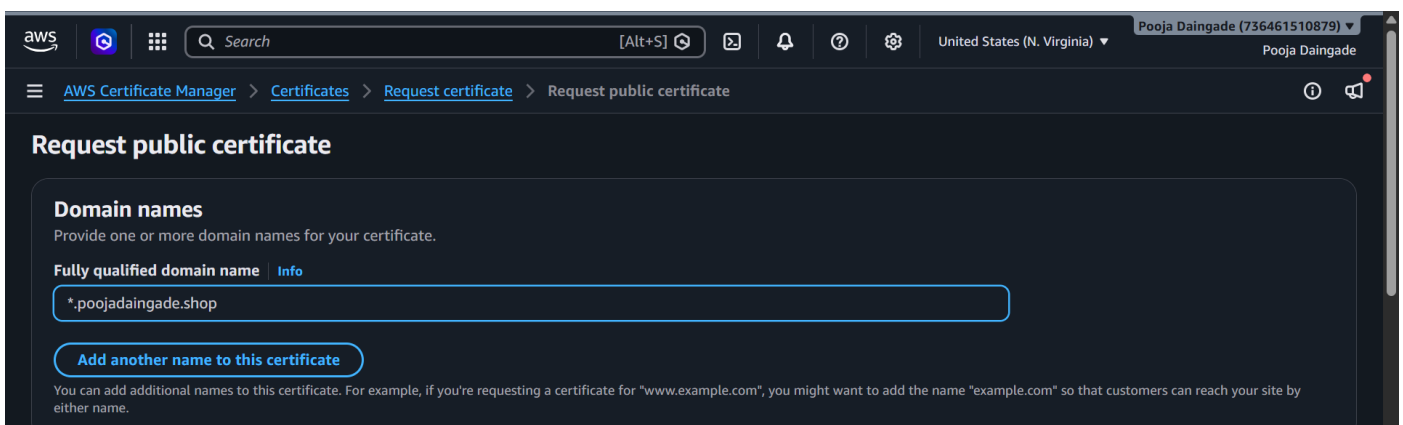
3. Uploaded the static website files, including **index.html** and **style.css**, to the **pooja-static-web-bucket** S3 bucket. Verified that both files were successfully uploaded and are available in the bucket.



4. Request an SSL/TLS Certificate using ACM → Selected **Request a public certificate** in AWS Certificate Manager (ACM) to secure the website with HTTPS. The certificate was requested in the **US East (N. Virginia) – us-east-1** region for CloudFront integration.



Added ***.poojadaingade.shop** as the domain name for the SSL certificate.



5. The ACM certificate status was **Pending validation**. Clicked **Create records in Route 53** to automatically create the required **CNAME DNS validation record** in Route 53.

The screenshot shows the AWS Certificate Manager console for a certificate with ID **c099fdee-33d1-4c68-b7bb-eaa7abdd7de3**. The certificate status is **Pending validation**. Below the status, there is a table of domains. The domain ***.poojadingade.shop** is listed with a status of **Pending validation** and a CNAME name of **_70cf1acf9e8d612363a0fdee073f5dac.poojadingade.shop**. A button **Create records in Route 53** is visible in the top right corner of the domains section.

Domain	Status	Renewal status	Type	CNAME name	CNAME value
*.poojadingade.shop	Pending validation	-	CNAME	_70cf1acf9e8d612363a0fdee073f5dac.poojadingade.shop	_7ee2269d34a3a75945b5d728e85afc85.jkddztszm.acm-validations.aws

The screenshot shows the 'Create DNS records in Amazon Route 53' wizard. It displays a table with the domain ***.poojadingade.shop** and its validation status **Pending validation**. The wizard includes a search bar for domains and a 'Create records' button at the bottom right.

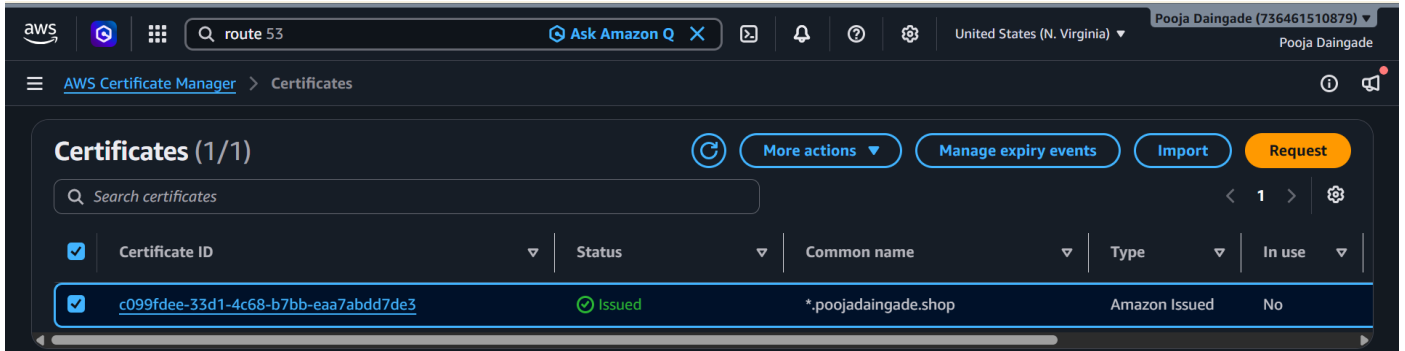
Domain	Validation status	Is domain in Route 53?
*.poojadingade.shop	Pending validation	Yes

6. Verified that the **CNAME validation record** was successfully created in the Route 53.

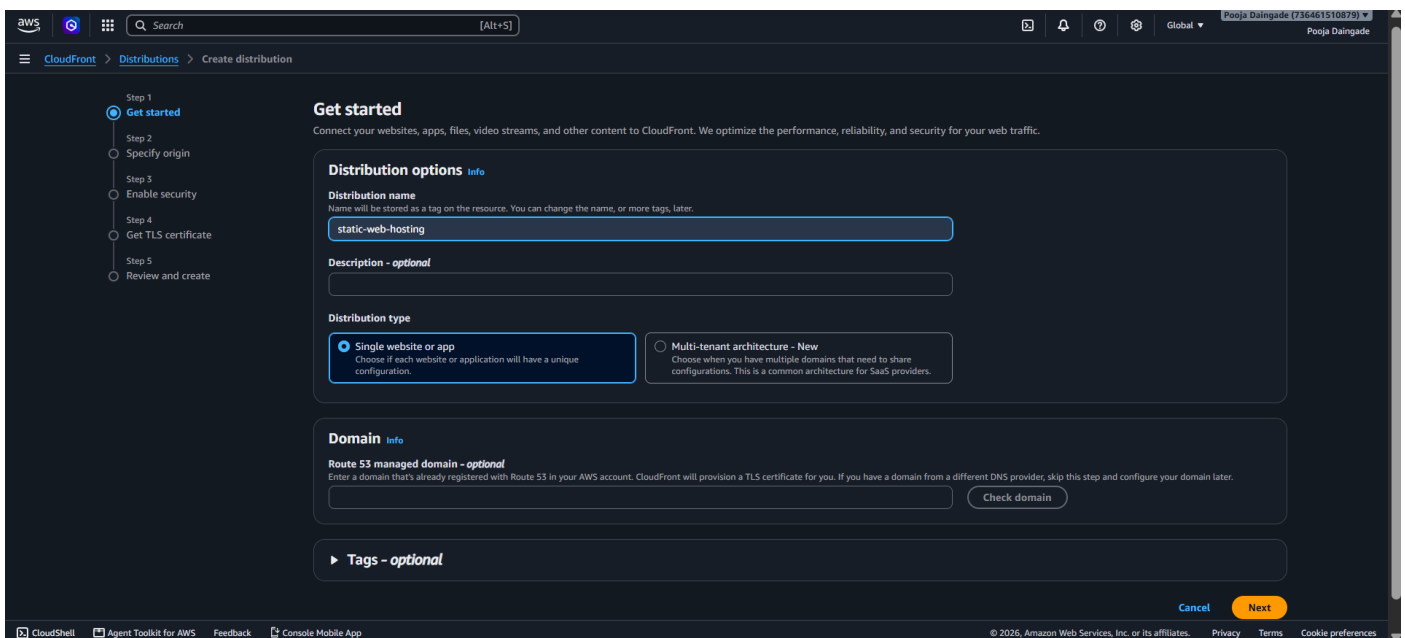
The screenshot shows the AWS Route 53 console for the hosted zone **poojadingade.shop**. The 'Records' tab is selected, showing a list of records. The CNAME record **_70cf1acf9e8d612363a0fdee073f5dac.poojadingade.shop** is highlighted, showing its value **_7ee2269d34a3a75945b5d728e85afc85.jkddztszm.acm-validations.aws** and TTL of 300 seconds.

Record	Type	Routing policy	Alias	Value/Route traffic to	TTL (seconds)	Health check	Evaluate health
poojadingade.shop	NS	Simple	No	ns-915.awsdns-50.net, ns-31.awsdns-03.com, ns-1541.awsdns-00.co.uk, ns-1150.awsdns-15.org	172,800	-	-
poojadingade.shop	SOA	Simple	No	ns-915.awsdns-50.net. awsdns-50.org.	900	-	-
_70cf1acf9e8d612363a0fdee073f5dac.poojadingade.shop	CNAME	Simple	No	_7ee2269d34a3a75945b5d728e85afc85.jkddztszm.acm-validations.aws	300	-	-

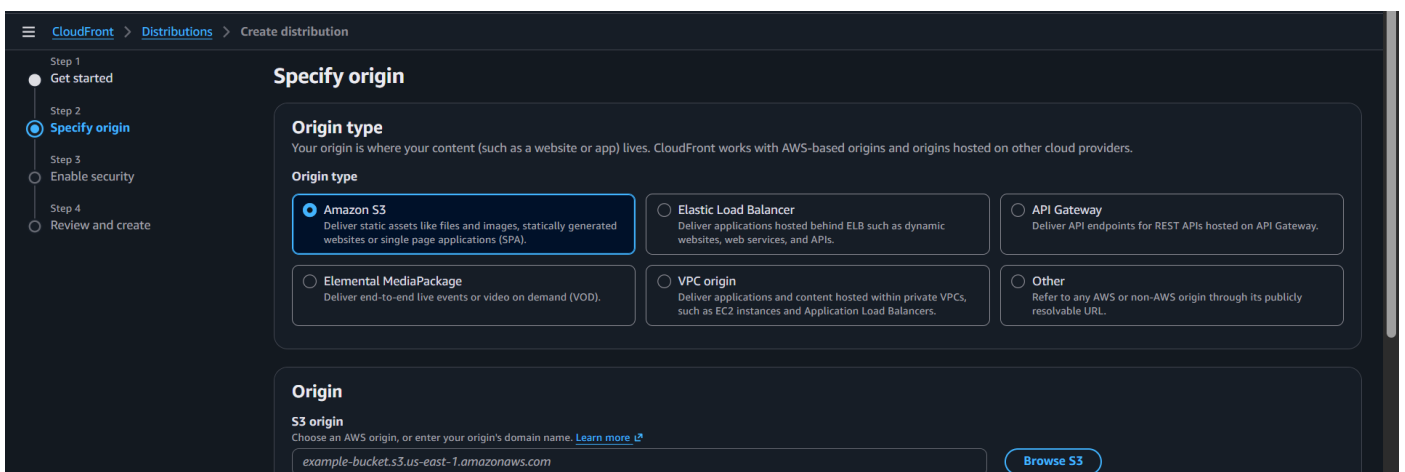
7. Verified that the **SSL/TLS certificate** status is **Issued**, confirming successful domain validation and certificate activation.



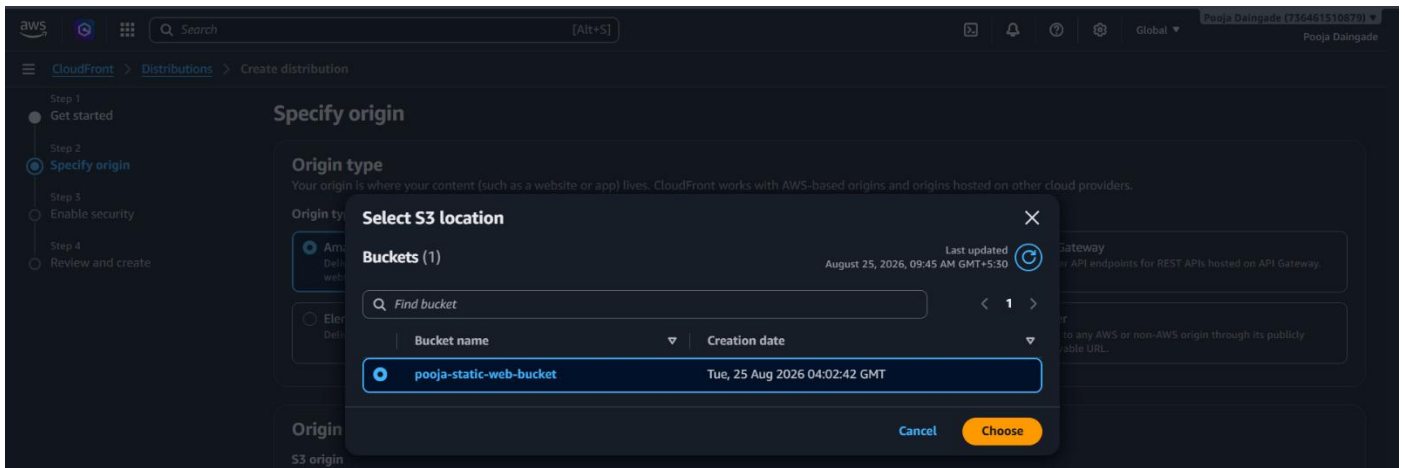
8. Created an Amazon CloudFront distribution named **static-web-hosting** to deliver the S3-hosted static website securely over HTTPS and clicked **Next** to continue the distribution configuration.



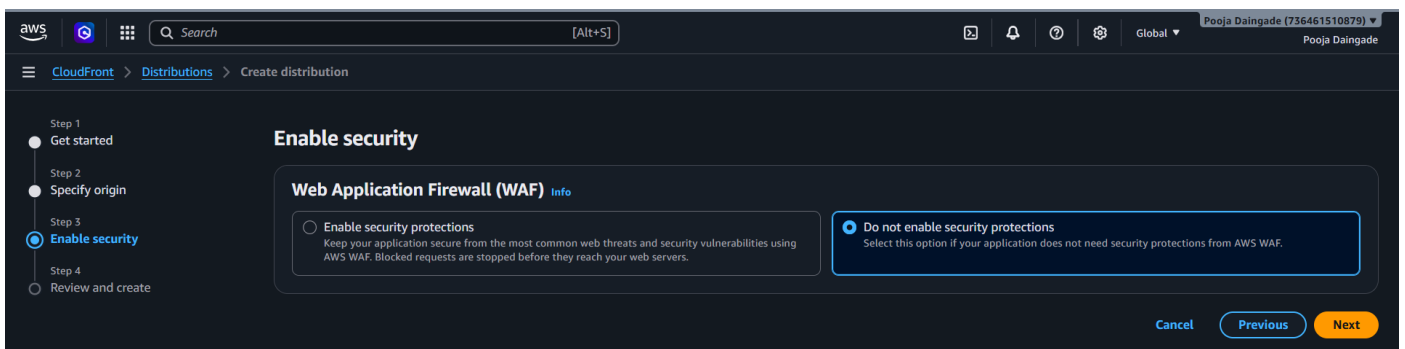
9. Selected **Amazon S3** as the origin type.



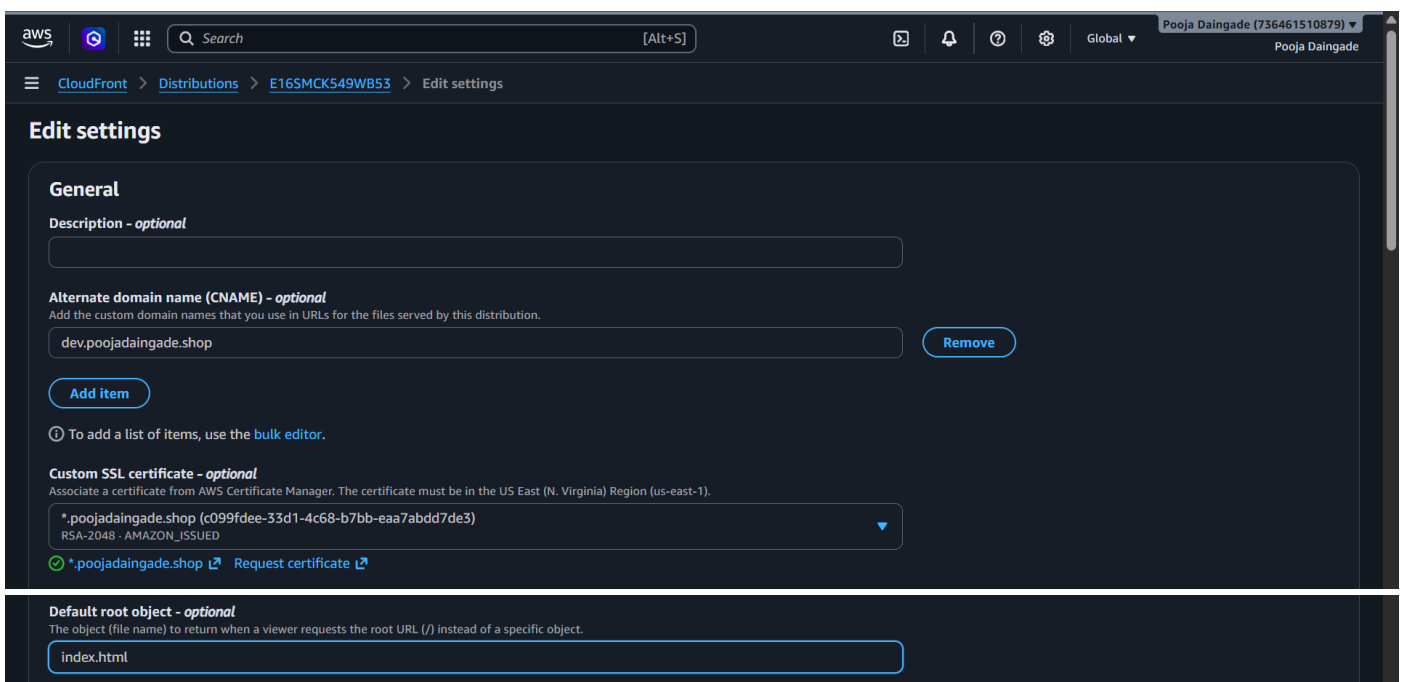
Clicked **Browse S3** to choose the **pooja-static-web-bucket** S3 bucket.



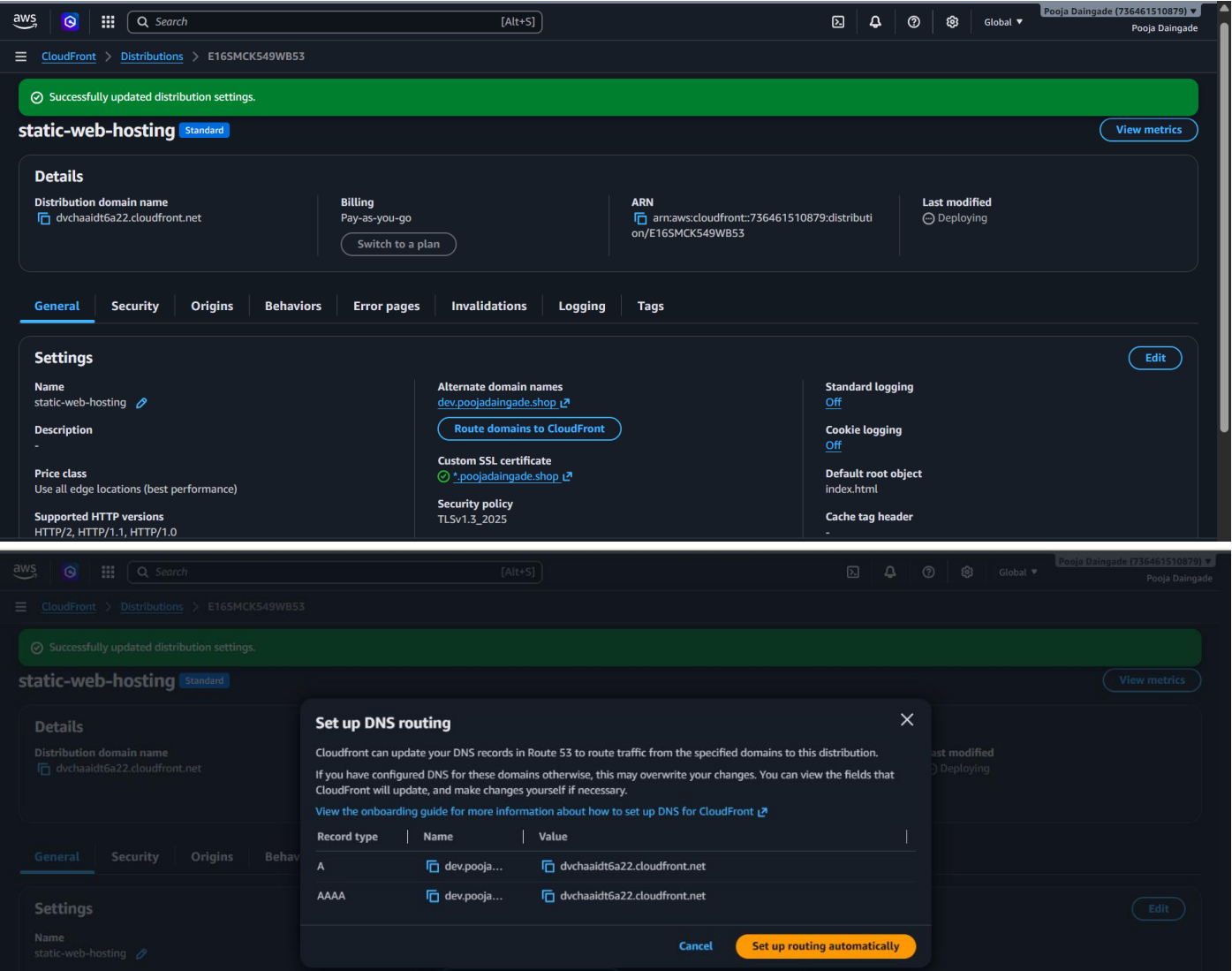
10. Selected **Do not enable security protections** and clicked **Create distribution**.



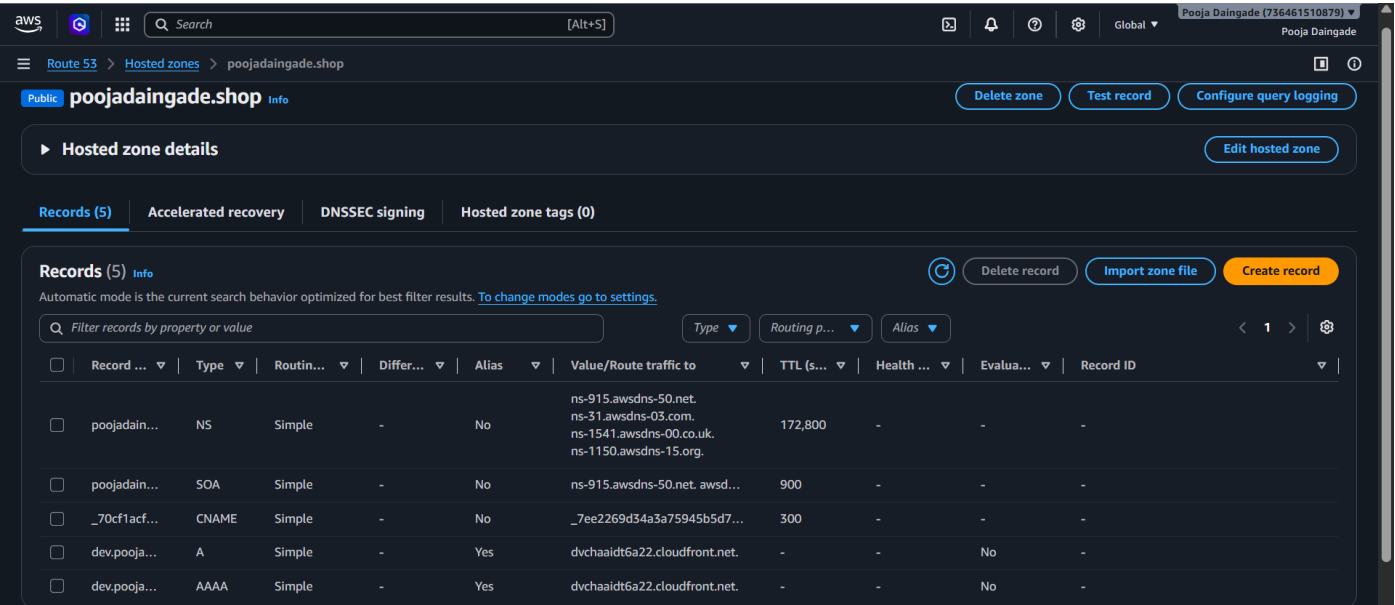
11. In CloudFront General Settings, clicked **Add item** to add the CNAME **dev.poojadaingade.shop** as an Alternate Domain Name. Selected the issued ACM SSL/TLS certificate for ***.poojadaingade.shop** to enable HTTPS. Set the Default Root Object to **index.html** and clicked **Save changes**.



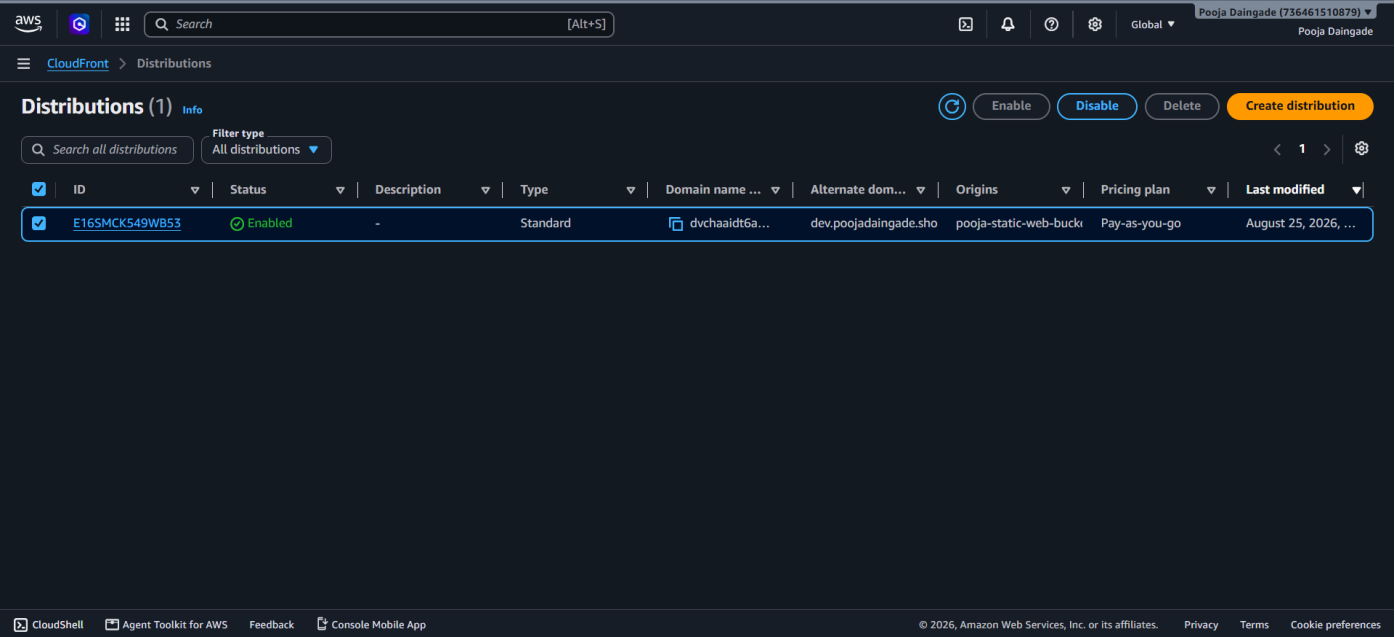
12. Selected **Route domain to CloudFront** and clicked **Set up routing automatically** to configure the Route 53 DNS record for the CloudFront distribution.



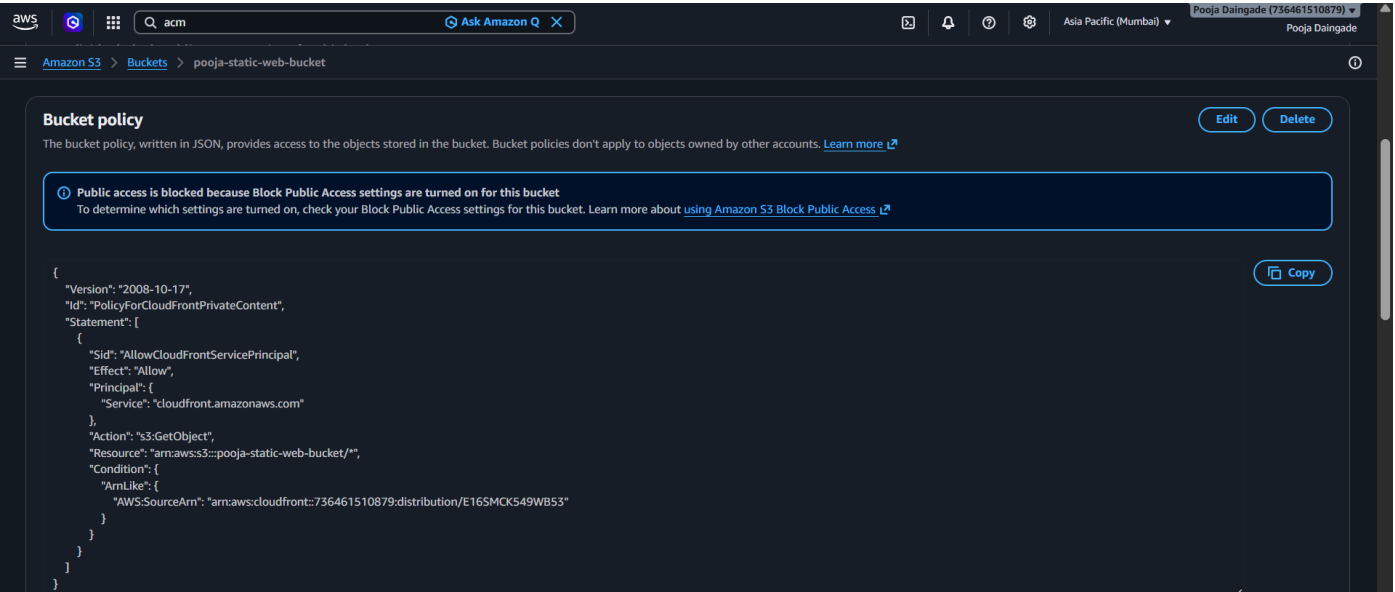
13. Verified that the **A and AAAA Alias records** were automatically created in Route 53 and configured to route dev.poojadingade.shop to the CloudFront distribution.



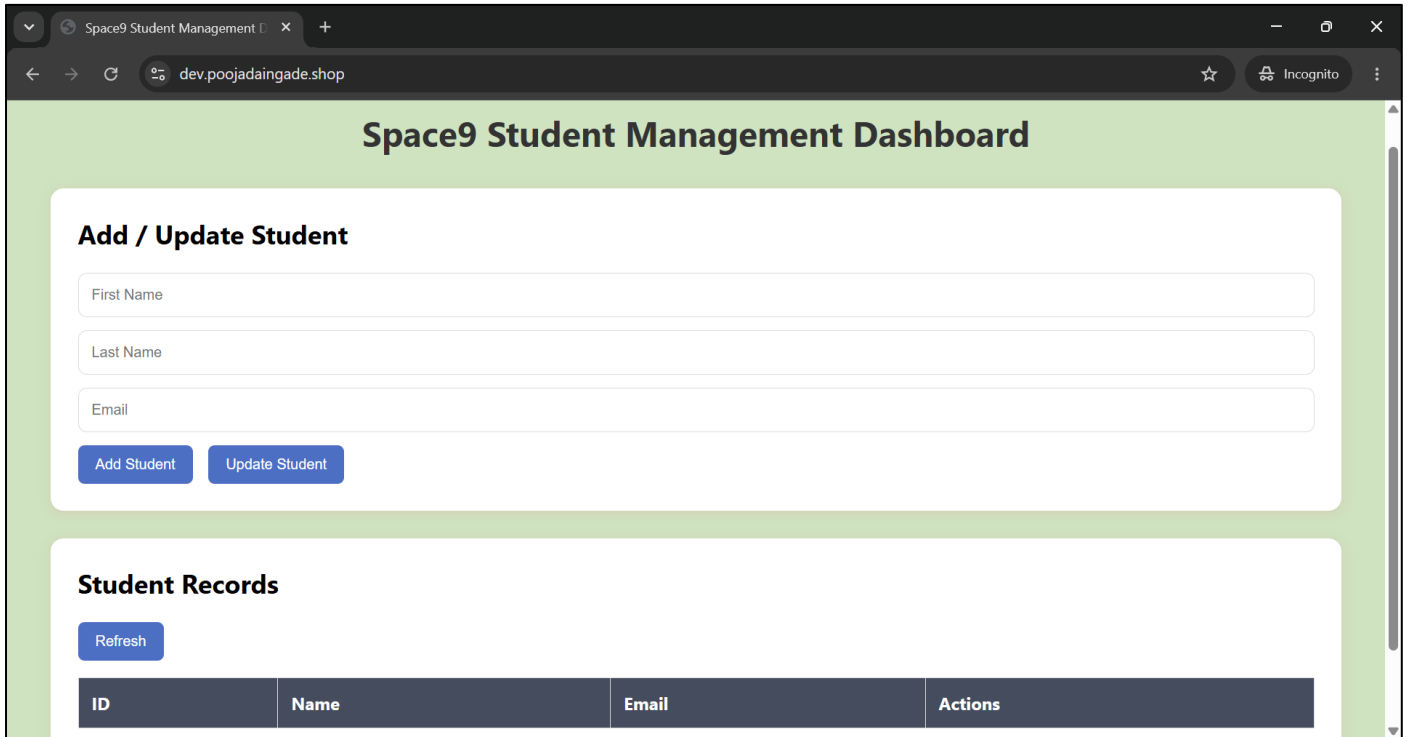
14. Verified that the **CloudFront distribution is successfully created and ready**, with the custom domain configured for HTTPS access.



15. Verified that the **CloudFront bucket policy** was automatically created to securely allow CloudFront to access objects from the private S3 bucket.



16. Successfully accessed <https://dev.poojadingade.shop> and verified that the static website is successfully deployed and securely accessible over **HTTPS** through CloudFront.



Conclusion

Successfully deployed and secured a static website using Amazon S3, Amazon CloudFront, AWS Certificate Manager (ACM), and Amazon Route 53. The custom domain was configured with an SSL/TLS certificate, enabling secure HTTPS access through CloudFront while keeping the S3 bucket private and protected.